BENJAMIN MBAU MUHOYA

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INTERESTS

I am broadly interested in how the environment shapes variation in metabolism and health among people. More specifically, my research focuses on cholesterol metabolism. I seek to understand the trade-offs between investment in cholesterol metabolism and immune activity, given that cholesterol metabolism is reprogrammed to support immune responses, and we don't yet fully understand how it is reshaped. To address this problem, I work with the Turkana people who live in Northern Kenya; notably, these people are spread across different locations with varying environmental exposures. Some people still practice nomadic pastoralism in the more remote, arid regions, while others have embraced market-integrated lifestyles typical in post-industrial societies.

EDUCATION

PRINCETON UNIVERSITY [CLASS OF 2027]

- **Ph.D. Ecology and Evolutionary Biology** (Currently in my 6th Year)
    - **Relevant Coursework:** Fundamental Concepts in Ecology, Evolution, & Behavior, Responsible Conduct in Research, Tropical Ecology, Colloquium on the Biology of Populations, Statistics for Social Science, Introduction to Chemistry Instrumentation, Programming for Biologists, Statistics for Biologists.

TECHNICAL UNIVERSITY OF KENYA [CLASS OF 2016]

- **Bachelor's in Medical Laboratory Science (BSc. MLS):**
    - **Relevant Coursework:** Human Genetics, Recombinant DNA Techniques, Bioinformatics, Biostatistics, Systemic Microbiology, Cell Biology, Molecular Diagnostics, Molecular Physiology, Community Health, Research Methods, and Bioethics.
    - **Senior Year Project:** Prevalence and sexual dimorphisms of Colon Cancer in Nairobi Hospital and Kenyatta National Hospital

RESEARCH EXPERIENCE

Mpala Research Center, Kenya (June 2023-August 2023)

Second Year summer field season with the Turkana Health and Genomics Project (THGP)

Committee: Dr. Andrea Graham, Dr. Jessica C. Metcalf, Dr. Julien Ayroles, Princeton EEB, and Dr. Paivi Pajukanta, UCLA.

Activities: Collecting additional samples to increase the power of my analysis of genomic regions influencing metabolic traits among the Turkana pastoralist community.

Mpala Research Center, Kenya (June 2022-August 2022)

First Year summer field season with the Turkana Health and Genomics Project (THGP)

Committee: Dr. Andrea Graham, Dr. Jessica C. Metcalf, Dr. Julien Ayroles, Princeton EEB, and Dr. Paivi Pajukanta, UCLA.

Activities: I did sample collection and processing (DNA and Serum) to generate data for a planned metabolomics experiment focused on the effects of lifestyle transition on metabolism among the Turkana pastoral community. I also helped train new field assistants to expand the project and facilitate data collection, including fine-tuned questionnaires that capture dietary information.

Mpala Research Center, Kenya (2018-2021)

Senior Research Assistant for the Turkana Health and Genomics Project (THGP)

PI: Dr. Julien Ayroles, Assistant Professor, EEB and LSI Evolutionary quantitative genomics

Activities: I facilitated sample collection and processing (DNA, RNA, serum, and PBMCs extraction) for cardiometabolic health research. I also performed a manual differential white blood cell count to assess the nature of the participants' immunity based on the WBC differential. We assessed the participants' blood sugar levels, blood pressure, and lipid profiles as part of research aimed at understanding the distribution of diabetes markers across the study populations. I also wrote detailed reports on the Prevalence of Prediabetes conditions among the Turkana Population.

The Nairobi Hospital & Kenyatta National Hospital, Kenya (2014-2015):

Undergraduate Senior year research.

Supervisor: Dr. Mercy Muriuki, lecturer at The Technical University of Kenya

Activities: I went through hospital record books and collected data on colon cancer cases. Following this, I did a statistical analysis using IBM SPSS Software and wrote my senior year thesis (age and sexual dimorphism of colon cancer cases: a case study on KNRH, and Nairobi Hospital), among the best projects in my undergraduate class, Class of 2016; I scored an A in the Research Methods Course.

PUBLICATIONS

1. Lea A. J., Caldas I. V., Garske K. M., Kahumbu J., Kinyua P., Miano C., Muhoya B., Peng J., Rabinowitz J. D., Roichman A., et al. Adaptations to water stress and pastoralism – Science (2025)
2. Lea, Amanda J., Charles Waigwa, Muhoya B., Francis Lotukoi, Julie Peng, Lucas Henry, Varada Abhyankar et al. Socioeconomic status effects on health vary between rural and urban Turkana— Evol.Med.(2021).
3. Arner, A. M., Muhoya, B., Snyder-Mackler, N., Ayroles, J. F., & Lea, A. J. (2024). Sex differences in immune function and disease risk are not easily explained by an evolutionary mismatch. bioRxiv (2024).
4. Watowich M. M., Arner A. M., Wang S., Mwai C. M., Muhoya B., Mukoma B., Wallace I. J., Ayroles J. F., Kraft T. S., Lea A. J. et al. The built environment is more predictive of cardiometabolic health than other aspects of lifestyle in two rapidly transitioning Indigenous populations – bioRxiv (2024).

TEACHING

2026 Princeton University EEB 347 "Tropical Forest Ecology" - Field Course at Barro Colorado Island, Panama

2025 Princeton University EEB 355 "Statistics for Biologists."

2024 Princeton University EEB 321 "Ecology: Species Interactions, Biodiversity and Society."

2023 Princeton University EEB388 "Genomics in the wild." - Field Course at Mpala Research Center, Kenya

PRESENTATIONS/POSTERS

1. Muhoya, Benjamin. *"Leveraging a natural lifestyle gradient to uncover hidden lipoprotein dynamics,"* Poster session at the Gordon Research Conference, New Hampshire, June 2026: Lipoprotein Metabolism.
2. Muhoya, Benjamin. *"Decoding how lifestyle shapes the Metabolome in a desert living pastoral community facing rapid Urbanization"* THGP Talks, Organized Zoom Meeting, August 2025.
3. Muhoya, Benjamin. *"Understanding Non-communicable Disease from an Evolutionary Mismatch Perspective."* Princeton University, Disease Group Talks, Organized Oral Session, March 2022.
4. Muhoya, Benjamin. *"Health shifts in urban Turkana are weakly mediated by increased consumption of market carbohydrates, and more strongly by indices of urbanicity."* Mpala Research Center's Science, Education, and Conservation Day, Organized Oral Session, February 2020.
5. Muhoya, Benjamin. *"Age and sexual dimorphism of colon cancer cases: a case study on KNRH and Nairobi Hospital."* Technical University of Kenya: presentation of senior-year project findings to the School of Health Sciences faculty, August 2016.

PROFESSIONAL WORK EXPERIENCE

MPALA RESEARCH CENTER – NANYUKI

Senior Research Assistant & Lab Technologist (April 2019 – August 2021)

Key responsibilities:

- Assessing samples from our recruited cohort to gauge their cardiometabolic health and the effects of different environments on health (rural vs. urban).
- Preparing and staining thin blood films for microscopic analysis to determine the nature and number of white blood cells in participants.
- Extraction of DNA, RNA, serum, and PBMCs, and preservation for subsequent analysis using various methods to address the broad research questions guiding the project.
- Fieldwork (planning and logistics), data analysis, writing reports, and the presentation of results.

KING DAVID HOSPITAL – NGONG BRANCH

Lab Technologist (August 2018 – March 2019)

Key responsibilities:

- Carried out sample analysis (microscopy, clinical chemistry, and used the full hemogram machine) for medical diagnostic purposes and reported the results to the doctor.

STRATUS MEDICAL IMAGING SOLUTIONS

Lab Technologist (November 2016- July 2017)

Key responsibilities:

- I prepared specimens and performed biochemical tests, including UECs (Urea, electrolytes & creatinine), to assess kidney and liver function, and I reported the results to the doctor.

KENYATTA NATIONAL HOSPITAL

Intern (April 2015 – September 2015)

Key responsibilities:

- Worked with and calibrated several automated machines (flow cytometer, clinical chemistry analyzers, biochemical analyzers).
- Assisted in several complex sample retrieval processes (e.g., Lumbar puncture & Fine Needle Aspirate) as part of learning.

THE NAIROBI HOSPITAL & CANCER TREATMENT CENTER

Intern (September 2014- December 2014).

Key responsibilities:

- Processed tissue samples by embedding them in wax, cutting thin sections using the microtome machine, and preparing thin tissue section slides for analysis by the senior pathologist (for cancer research).
- Prepared agar plates for bacterial culture, cultured bacteria, and read the plates following bacterial growth to facilitate diagnosis by the senior microbiologist. We performed antibiotic susceptibility tests on some cultured bacteria to aid the doctor's prescription.

PROFESSIONAL TRAINING

1. Wilderness First Aid (WFA) course at Princeton University: Through the course, I earned certifications in remote emergency response, prolonged patient care, and improvisation through outdoor simulations.
2. Cold Spring Harbor Lab (CSHL): Programming for Biology Course (Oct 2023): The course helped me develop valuable skills in computational techniques (programming in Python, Linux, and Unix) for handling large datasets, such as those generated in genomic research.
3. EEB Genomics Field Courses at the Mpala Research Center (2019): I learned to plan fieldwork, design experiments, and analyze field-collected data using RStudio.
4. Kenya Medical Laboratory and Technologists & Technicians Board (KMLTTB) – 2017. I attained a lab technologist's practice license from the Kenyan government.
5. St John's Ambulance Training: For certification in basic first aid skills.
6. Kenya Red Cross Community-Based First Aid Training: I completed a course on basic first aid skills, including CPR, emergency childbirth, managing fever, basic HIV/AIDS education, handling common diseases, and basic firefighting skills.